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MILITARY SPECIFICATION
TRUCK, AIRCRAFT, CARGO LOADING-UNLOADING, 25,000 POUNDS CAPACITY A/S32H-5

"This specification is approved for use within (Code 84) Department of the (Air Force), and is available for use by all Departments and Agencies of the Department of Defense."

1. SCOPE

1.1 This specification establishes the requirements for manufacture and acceptance of a self-propelled, adjustable height, 25,000 pound capacity aircraft loading and unloading truck.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTB, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the 2nd of this document or by letter.

AMSC N/A FSC 3930

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SPECIFICATIONS

FEDERAL

W-B-131	Battery, Storage (Vehicular, Ignition, Lighting and Starting)
QQ-P-416	Plating, Cadmium (Electrodeposited)
VV-F-800	Fuel Oil, Diesel

MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-W-5044	Walkway Compound, Nonslip, and Walkway Matting, Nonslip
MIL-C-5541	Coatings, Chemical Conversion on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-27443	Pallet, Cargo, Aircraft, HCU-6/E, HCU-12/E and HCU-10/C
MIL-P-44115	Platform, Airdrop, Type V
MIL-C-46168	Coatings, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-W-52574	Welding Process and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels
MIL-T-52864	Trucks, Lift, Fork: DED, Pneumatic Tires, 10,000, 15,000, and 20,000 Pound Capacity; General Specification For
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-C-62122	Cable Assembly, Slave, Electrical, Connectors, Plug and Receptacle
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-595	Colors

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MIL-STD-12	Abbreviations for Use on Drawings, and on Specifications, Standards and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-808	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-907	Engineering and Design Criteria for Shelters, Expandable and Nonexpandable
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1791	Internal Aerial Delivery in Fixed Wing Aircraft, Designing for

FOREST SERVICE, US DEPARTMENT OF AGRICULTURE

Standard 5100-1A	Spark Arrestors for Internal Combustion Engines

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues

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shall be those in effect on the date specified in the solicitation.

LAWS AND REGULATIONS

49CFR 393	Federal Motor Carrier Safety Regulations (FMCSR)
49CFR 571	Federal Motor Vehicle Safety Regulations (FMVSS)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

(Copies of specifications, standards, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are Department of Defense (DOD) adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be those in effect on the date specified in the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A123	Standard Specification for Zinc (Hot Dip Galvanized) Coatings on Iron and Steel Products
ASTM B633	Electrodeposited Coatings for Zinc on Iron and Steel, Specification for

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION

Standards

(Application for copies should be addressed to European Tyre and Rim Technical Organization, 32 Avenue, Brugmann-B, 1060 Brussells Belgium).

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 668	Freight Containers - Classification External Dimensions and Ratings
ISO 8323	Freight Containers - Air/Surface (Intermodal) General Purpose Containers - Specifications and Tests

(Application for copies should be addressed to American National Standard Institute, 1430 Broadway, New York NY 10018.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 407

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston MA 02110.)

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

(Application for copies should be addressed to Superintendent of Documents, US Government Printing Office, Washington DC 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J293	Vehicle Grade Parking Performance Requirements
J336a	Sound Level for Truck Cab Interior
J360	Motor Vehicle Grade Parking Performance Test Code
J366b	Exterior Sound Level for Heavy Trucks and Buses
J517	Hydraulic Hose
J534	Lubrication Fittings
J695	Turning Ability and Off Tracking - Motor Vehicles
J786a	Brake System Road Test Code - Truck, Bus and Combination of Vehicles
J821	Electrical System for Construction and Industrial Machinery
J992b	Brake System Performance Requirements - Truck, Bus and Combination of Vehicles
J994	Alarm - Backup - Electric - Performance, Test and Application
J1077	Measurement of Exterior Sound Level of Trucks with Auxiliary Equipment
J1349	Engine Power Test Code, Spark Ignition Diesel

(Application for copies should be addressed to Society of

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Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

TIRE AND RIM ASSOCIATION

Yearbook

(Application for copies should be addressed to Tire and Rim Association, Incorporated, 3200 West Market Street, Akron OH 44313.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type document. Except as otherwise specified, wood shall be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so

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that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, removal and replacement.

3.1.3.1 Reliability. The vehicle, including all components, shall have an observed Mean-Time-Between-Failures (MTBF) of not less than 100 hours, demonstrated in accordance with (see 4.6.7). A failure is defined as the event or inoperable state in which any item, or part of an item, does not, or would not, perform as previously specified. Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-STD-781. In the event of a conflict between those documents, MIL-STD-781 shall take precedence. The vehicle shall be designed for a life expectancy of at least 12 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than two hours.

g. Maximum corrective maintenance downtime of no greater than eight hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

3.1.3.2.1 Maintenance tasks. The vehicle shall be designed so that one mechanic using common hand tools can perform the following maintenance operations within the time specified:

a. Remove, replace and adjust all engine driven belts - 3/4 hour.

b. Remove and replace alternator - 1/2 hour.

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c. Remove and replace all filters, screens and strainers in all hydraulic systems - one hour.

d. Remove and replace engine coolant system hoses - 1/2 hour.

e. Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase - one hour.

f. Remove and replace fuel filter element(s) - 1/2 hour.

g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1/2 hour.

h. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - one hour.

i. Remove and replace starter - one hour.

j. Bleed and adjust brakes and refill master cylinder (if applicable), two mechanics - 3/4 hour.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.3.2) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle

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shall be free of hazardous protrusions, sharp edges or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.4 Environmental conditions. The vehicle shall be capable of satisfactory storage and operation as described herein under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -40° F to +125° F for operation; -60° F to +160° F for storage.

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 mm to 4.5 mm, falling at a rate of 13 cm per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0ml/80cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750 ft/min, concentration 0.5g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 7,500 ft/min, concentration 0.0773g/ft^3.

3.1.5 Weight distribution. The weight distribution of the vehicle in the unloaded condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed the component manufacturer's recommendations.

3.1.6 Vehicle lean. The vehicle fully loaded or unloaded shall not lean to either side more than three percent of the maximum width when the deck is positioned in the travel mode. Vehicle lean measurements shall be made at the outermost portion of the vehicle (excluding mirrors, mirror support brackets, lights or asymmetrical shapes) directly above each axle; the difference in vehicle height at each axle shall not exceed three percent of the maximum width at that location.

3.1.7 Electromagnetic interference. The vehicle shall comply

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with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for test UMO3 (Figure 8-1) shall be relaxed by 20dB.

3.1.8 Sound levels.

3.1.8.1 Exterior sound level. The maximum sound level produced by the vehicle shall not exceed 83dBA when measured in accordance with SAE J366b. An additional 2dBA allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment and inherent differences in nominally identical vehicles.

3.1.8.2 Cab interior sound level. The maximum sound level in the cab interior shall not exceed the SAE recommended design criteria for new vehicles when measured in accordance with SAE J336a. Except select a transmission and/or axle gear ratio so that approximately 15 MPH is obtained at rated engine speed.

3.1.8.3 Equipment operation sound level. The maximum sound level produced by the vehicle shall not exceed 85dBA when measured in accordance with SAE J1077. In addition, the vehicle shall be positioned so that the microphone is located 24 inches from the auxiliary equipment operator's control panel when tested by the stationary test procedure.

3.1.9 Air transportability. The vehicle shall be air transportable without the use of shoring in accordance with MIL-STD-1791 in Types C-130, C-141 and C-5 Aircraft and shall be air transportable in Type C-17 Aircraft. Preliminary design guidelines for Type C-17 are 142 inch maximum height, 204 inch maximum width, 812 inch maximum length, 9 degree ramp angle and floor load capacity equal to that allowed for the Type C-5 Aircraft. Two rows and two layers of 2 inch x 12 inch finished lumber shoring is allowed in the C130 aircraft at the rails to support the wheel loads.

3.1.9.1 Equipment removal. Removal of equipment shall be permitted, if necessary, to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies, which must be handled or positioned, shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall take not longer than 30 minutes for two mechanics using common hand tools. While being driven and steered from the operator's cab, the truck shall negotiate up and down the aircraft cargo ramps without the use of shoring.

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3.1.9.2 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.1.10 Commercial components. Commercial components shall be used where possible and shall not be used outside their commercial ratings.

3.1.11 Structure. Based on yield strength, the structures (deck, lift mechanisms and chassis), including the guide rails, pallet restraints and emergency pallet stops, shall be designed to provide a static safety factor of three, or dynamic safety factor of two while accomplishing any of the performance requirements described herein, whichever is greater. The load criteria shall be the same as that described (see 3.8).

3.2 Performance.

3.2.1 Mobility. The fully loaded and equipped vehicle in the travel mode shall:

a. Maintain a speed of 15 miles per hour (MPH) on a level paved road.

b. Accelerate from a stop to a speed of 15 MPH within 30 seconds on a level paved road.

c. Be capable of traversing a three percent grade in a tractive environment resulting from rain, snow, sleet, ice, sand and mud on a prepared surface.

d. Negotiate a 5 percent side slope in each direction.

e. Negotiate, in each direction, a ramp with a 15.0 degree slope connecting two horizontal surfaces without spillage of any fluid when configured for air transport. The ramp length shall be 20.0 feet. The vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascend or descend the ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the vehicle, other than its tires, contacting any of the three surfaces.

f. Sustain operation over paved surfaces of three percent grades at a forward speed of 12 MPH and a reverse speed of 5 MPH.

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g. Traverse Belgian Block Course (see 4.6.3.8) at a speed of 4 MPH.

h. Top vehicle speed shall be not more than 25 MPH.

i. Comply with all service brake performance requirements of SAE J992b.

j. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 20.4 percent.

k. Have a wall to wall turning diameter of 100 feet maximum in either direction.

l. Be capable of withstanding repeated impacts at front and rear bumpers with a permanent loading dock at a speed of 2.5 MPH.

m. Ascend and descent a 30 percent grade without spillage of any fluid.

n. Traverse graded gravel and dirt roads at a speed of 10 MPH.

o. Traverse, in one pass at low speed, terrain consisting of unflooded fine grained dry to moist soils having a minimum cone index rating in the critical layer of no greater than 90. The vehicle cone index rating shall be measured in accordance with the Waterways Experimental Station Mobility Index Method.

3.2.2 Axle articulation. Sufficient articulation shall be provided to ensure that loading on each wheel does not exceed the rating of the wheel manufacturer when any one wheel is elevated five inches higher above the horizontal plane than the other wheels. The axle shall articulate sufficiently to allow the loader to negotiate the transporting aircraft loading ramp. Vehicle shall be capable of negotiating 3 1/2 inch high obstacles spaced to provide a racking of the chassis at 4 MPH. Axle travel shall be limited to preclude interference damage with other vehicular components.

3.2.3 Deck operation. When truck is loaded as specified (see 3.8) it shall be capable of performing the following exercise cycle repeatedly while maintaining a hydraulic system fluid temperature below 185° F. One exercise cycle shall consist of the following:

a. Lift chassis to upper limit.

b. Lower chassis to lower limit.

c. Lift deck to upper limit.

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d. Roll deck full left, full right and return to level.

e. Pitch full forward, full rear and return to level.

f. Side shift deck full left, full right and return to center.

g. Lower deck to bottom limit.

3.3 Chassis.

3.3.1 Powertrain.

3.3.1.1 Engine. The vehicle shall be equipped with a commercial liquid cooled diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624 and Grade JP-8 in accordance with MIL-T-83133. Any engine that is adversely affected by any of the above fuels is not acceptable. The engine governed speed shall not exceed the engine manufacturer's normal commercial recommendations.

3.3.1.1.1 Engine installation. The engine shall be installed to allow easy access for maintenance operations (see 3.1.3.2). Engine and transmission shall be dampened so as not to adversely affect other vehicle components.

3.3.1.1.2 Engine lubrication system. Lubrication to all internal engine components shall be provided during all vehicle attitudes and mobility requirements (see 3.2.1). The oil filler opening shall be accessible with a five quart oil container so as to allow replenishment of engine oil.

3.3.1.1.3 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided.

3.3.1.1.4 Engine starting system. The engine shall start within 5 minutes in any ambient temperature within the required operating range of the vehicle. Installed glow plugs, fluid starting aids and heat from the winterization system (see 3.6) may be used prior to and during the start period to facilitate engine starting under the following conditions:

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TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Glow plugs and fluid starting aids
-40 through 0	Glow plugs, fluid starting aids and heat from the winterization system

3.3.1.1.4.1 Start interlocks. The starting system shall be equipped with an interlock device to prevent starter engagement when the engine is running. A safety device shall be installed to prevent starting of the engine in any transmission selector setting other than neutral. Interlocks shall be provided to prevent starting of engine when parking brake is disengaged and/or selector control (see 3.4.7) is positioned at other than automotive function.

3.3.1.1.5 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain.

3.3.1.1.6 Fuel depletion. Fuel depletion conditions shall be corrected by a simple, accessible fuel priming and engine restart system procedure.

3.3.1.1.7 Spark arrestor. A spark arrestor conforming to Forest Standard 5100-1A shall be furnished.

3.3.1.1.8 Emergency shutdown. A cab located emergency shutdown control shall be provided. The "kill switch" shall be within easy reach of the loader operator. Actuation of this control shall be a means of stopping all engine functions, systems and movement in the event of failure of normal controls.

3.3.1.2 Transmission. A commercial (see 6.3.2) automatic transmission shall be provided, complete with a hydraulic torque converter. The transmission shall have a torque capacity of at least 110 percent of the maximum net engine torque. The transmission shall be capable of infinitely variable speeds as it approaches the aircraft for load transfer.

3.3.1.3 Axles. Commercial (see 6.3.2) axles shall be provided. The load rating of each axle shall be not less than imposed load

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on the axle measured at the ground with the vehicle at gross weight. The number of axles shall be governed by the gross vehicle weight, center of gravity, air transportability and vehicle dynamics.

3.3.2 Steering system. The vehicle shall be equipped with a power steering system and shall provide steering control in the event of power failure. The steering system shall not require more than four turns lock to lock.

3.3.2.1 Steering wheel. The steering wheel shall have a diameter of 18 inches minimum. Not less than four inches clearance shall be provided around the steering wheel in all directions.

3.3.3 Brakes.

3.3.3.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. Brakes shall be of either drum or disc type. A commercial (see 6.3.2) air dryer and necessary associated parts shall be added to the pneumatic system to eliminate excessive moisture and oil, thus increasing the life of the air brake components. The air dryer shall use desiccant for the final absorption process and shall be installed in the discharge line between the compressor and the first reservoir. A heater and thermostat assembly shall be provided with the air dryer to preclude freeze-up when used in severe winter conditions. A petcock drain shall be provided at the lowest point on each air reservoir and shall be readily accessible for service from the side of the vehicle. At least two air reservoirs, one wet and one dry, shall be furnished.

3.3.3.2 Parking brakes. The vehicle shall be equipped with parking brakes of the spring applied type on all nonsteering wheels.

3.3.4 Frame.

3.3.4.1 Tow hooks. Two tow hooks shall be provided on the chassis front and rear for towing the loader to the maintenance shop if required.

3.3.4.2 Stability. Track and wheelbase dimensions shall be such that optimum stability of loader is achieved. Under the following combined conditions, the truck shall be stable; meaning, loader structure movement shall be not more than 4.0 inches measured laterally and longitudinally along the deck top surface:

a. Speed up to 1.0 MPH in both forward and reverse directions.

b. Full turn, right and left hand.

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c. Deceleration rate, 12 feet/second^2.

d. Fully loaded.

e. Level road surface.

f. Deck elevated to top position.

g. Forty knot wind from any direction.

3.3.4.3 Maintenance supports. Maintenance supports capable of supporting the unloaded deck shall be attached to the truck for use during maintenance of the vehicle.

3.3.5 Suspension system.

3.3.5.1 Hydraulic actuation. A hydraulically actuated suspension system shall be provided. It shall be possible to adjust the suspension height in order that adequate clearance during road travel and ramp negotiation, and the minimum deck height requirement can be achieved. Damper(s) shall be furnished to minimize shock loads during road travel. If an accumulator(s) is used, it shall be bladder type.

3.3.5.2 Ground clearance. Road travel ground clearance shall be not less than 7 1/2 inches.

3.3.6 Wheels and tires. The vehicle shall be equipped with wheels and tires which meet all Tire and Rim Association or European Tyre and Rim Technical Organization requirements for this application. Tubeless steel belted radial ply tires with all terrain type tread shall be provided.

3.3.7 Fuel system.

3.3.7.1 Fuel tank. A fuel tank shall be provided with a capacity of not less than 30 US gallons sufficient to provide at least eight hours of continuous loader operation. The tank shall be a fill opening of three inches minimum, readily accessible to personnel standing on the ground. It shall be located and mounted so as to provide maximum protection from damage.

3.3.7.2 Fuel filter. At least one filter of the fuel water separator type shall be located in the fuel line between the fuel tank and the engine. It shall be installed so that the element can be readily serviced without removing surround equipment. Unless the filter is mounted higher than the fuel tank, a shut-off valve shall be included in the inlet line to the filter.

3.3.8 Air system. The air system shall provide and control power for the brake system, engine throttle and transmission shift controls.

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3.3.9 Backup alarm. An audible alarm shall be provided to caution personnel in the area when the vehicle is in reverse gear. The alarm shall be in accordance with SAE J994.

3.4 Cab. The cab shall swing or slide inward for air transport so that detachment is not required. The cab shall be mounted on the forward left (road side) side of the deck. The cab height above the loader deck shall be within 32 inches. The cab shall comply with all OSHA recommendations not precluded by the requirements of this specification. All deck and automotive functions shall be controlled from the cab with the exception of the emergency standby pump (see 3.7.7) and the emergency check valves (see 3.7.3).

3.4.1 Cab workspace. Workspace clearance and dimensional cab features shall be designed in accordance with MIL-STD-1472.

3.4.2 Seating. The operator's seat shall be designed following the recommendations of MIL-STD-1472 and FMVSS Number 207. Seating shall provide comfort by preventing fatigue stresses induced by long working/driving periods, oscillation and vibration resulting from hard chassis suspensions.

3.4.3 Doors. A door shall be provided on the left hand side of the cab to provide operator access in accordance with MIL-STD-1472, and the door locks and retention components shall be in accordance with FMVSS Number 206.

3.4.4 Cab egress/ingress. Easy access to the deck through the left hand side door shall be provided by a walkway or step(s) to allow safe exit/entrance. Hand restraints shall be provided as necessary.

3.4.5 Windshield and windows. Windows shall be installed in accordance with MIL-STD-1472. The right side window shall be a sliding type, and the left side (door side) window shall be roll type. The top window shall provide the operator with an overhead view of the area above the deck and cab. A protective screen shall be installed over the top window that will withstand a 200 pound load without permanent deformation to the window or the screen. The windows and windshield shall provide a lateral visual field or at least 180 degree and a ground view of all distances beyond six feet in front of the vehicle. The operator's line of sight shall be adequate to provide full view of the front end of the loader deck and aircraft opening during aircraft interface.

3.4.5.1 Windshield wipers and washers. Windshield wipers and washers shall be provided in accordance with FMVSS Number 104.

3.4.6 Mirrors. An adjustable rearview mirror shall be mounted on the left side of the vehicle on a retractable bracket.

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Rearview mirrors shall be in accordance with FMVSS Number 111.

3.4.7 Instruments and controls. Instruments and controls shall be located within the limits specified in MIL-STD-1472, arranged so that those which are most vital to operation have the most predominant location. They shall be placed in blocks distinctly marked to group those instruments and controls pertinent to the operation of any individual component. For example, all instruments and controls relative to the operation of the engine shall be in one block. Controls having similar functions shall have similar actions. Indicator lights shall be green to indicate proper operation and red when warning is intended. All instruments and controls shall be illuminated with nonglare red lighting to be readily visible for night operation. At least the following instruments and controls shall be provided.

a. Speedometer and odometer.

b. Tachometer and hourmeter.

c. Engine oil pressure gauge.

d. Engine coolant temperature gauge.

e. Transmission oil temperature gauge.

f. Voltmeter or ammeter.

g. Fuel level gauge.

h. Air pressure gauge.

i. Low air pressure warning light and buzzer.

3.4.7.1 Deck and drive controls. The control panel shall be marked with etched metal labels. Silk screen process is not allowed. All deck functions and suspension height adjustments shall be controlled by electrical command switches from the cab and shall be the "deadman type." The deck system shall be designed to minimize lag time between control command and deck function. A mode selector control with two positions labeled "drive" and "deck" shall be provided to preclude simultaneous deck and drive operations. When the truck is set in "deck" mode, the accelerator pedal shall be used to set the engine speed and shall allow the operator to adjust the engine speed for load variations. Suspension control shall be provided in both the "drive" and "deck" modes. Control shall provide for simultaneous actuation of any two deck functions. Interlocks shall be provided to require transmission to be in neutral and parking brakes to be engaged for operation of deck functions. Before vehicle can be placed in motion, the selector control must be set at the "drive" position. Cab shall be equipped with indicator

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lights showing when the deck is off center for the roll, pitch and side shift functions. Indicator lights shall be provided for the various suspension positions.

3.4.8 Climate control system. The cab shall be equipped with a diesel heater and sufficient insulation to maintain the cab air temperature at the floor above 40° F when the ambient temperature is below -40° F with no solar input. The heater shall include fresh air intakes, and discharge outlets shall be provided to direct heated air to floor and to windshield defroster louvers. The defroster shall be able to clear the windshield when exposed to a thin film of ice and at an ambient temperature as low as -40° F in less than 60 minutes and shall maintain clear vision. The heater shall be complete with blower and controls mounted convenient to the operator.

3.5 Electrical system. The vehicle shall be equipped with a 12 volt electrical system; a 24 volt starting system shall be provided. Wiring shall be weatherproof and color coded or numbered for easy identification in accordance with SAE J821. All circuits shall be protected by circuit breakers located so as to be protected from the environmental conditions described herein and to be readily accessible to a mechanic. A master switch shall be provided to disconnect the batteries from the electrical system. All electrical switches shall be failsafe.

3.5.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all vehicle electrical components simultaneously. Minimum capacity of the alternator shall be 130 amperes.

3.5.2 Batteries. The vehicle shall be equipped with two 12 volt batteries, rated at a total of 1,800 Cold Cranking Amps (CCA) minimum. The batteries shall comply with the maintenance free characteristics of W-B-131. Batteries shall be so arranged and connected to provide sufficient capacity for engine starting in all conditions specified herein. The batteries shall be mounted in an acid resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for servicing or replacement. An auxiliary power receptacle, MIL-C-62122, shall be installed near the batteries to permit engine slaving and battery charging and shall allow connection of auxiliary power cables by personnel while standing on the ground at the side of the chassis and with deck in any position.

3.5.3 Lighting.

3.5.3.1 Vehicle lighting. The vehicle shall be equipped with headlights, taillights, backup lights, parking lights, clearance lights, brake lights, turn signal lights and reflectors in accordance with Federal Motor Carrier Safety Regulations. A spotlight shall be mounted on the exterior of the cab and oper-

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able from the operator's seat for lighting the front half of the loader deck and the aircraft interface. The turn signal operating units shall have visible and audible flash indicators; turn signals shall be self-cancelling. Lights and reflectors mounted on the vertical surface of rub rails or vehicle bumpers shall be recessed.

3.5.3.2 Interior lighting. The cab and all compartments shall be provided with lighting adequate to enable personnel to readily obtain equipment and perform all functions necessary during night operations. Sealed contact switches shall be provided to automatically illuminate the lights in the cab or a compartment when its door is opened.

3.6 Winterization system. The vehicle shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -40° F. Heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power, plus one ground) male connector at the other end. Electrical insulation of connecting cable shall conform to FMCSR 393.77(c) (7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat, with an operating range of 140° to 160° F, shall be installed.

b. Immersion type engine oil heater, 300 watt minimum rating, shall be installed in oil pan though any convenient opening. The oil heater shall be controlled by the coolant heater thermostat.

c. Battery heater shall be provided, having a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10° F during vehicle exposure in ambient air temperatures as low as -40° F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80° F.

d. A fuel warmer or preheater shall be provided to prevent

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clogging of duel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40° F to an outlet temperature of +9° F, with a flow rate not less than the maximum fuel demand of the engine fuel system.

3.7 Hydraulic system. The hydraulic system shall be an "open center system"; i.e., pump runs constantly, open center valves allow return of hydraulic fluid back to the reservoir when not required to operate a function. The hydraulic system shall provide and control power for, but not limit to, roll, side shift, pitch, lift, conveyor, steering and suspension functions. The system shall consist of at least two discrete circuits. Each circuit shall include a pump or dual element which shall supply power to only that circuit under normal operating conditions and shall be capable of supplying power to any other circuit without removal of components should other circuit pump(s) fail. Pumps shall be identical or dual element type. Overload protection shall be provided on the high pressure lines from the hydraulic pumps to prevent damage to the hydraulic controls. Pressure relief valves shall be included at all hydraulic cylinders and motors to allow bypass of excess hydraulic pressure. The hydraulic supply shall have a filter system to insure delivery of clean fluid to the hydraulic circuit. The hydraulic system shall provide adequate flows and pressures so that simultaneous hydraulic actuation of any two deck functions shall not result in performance degradation.

3.7.1 Reservoir. The hydraulic fluid reservoir shall be designed to allow easy access to the fill port. The fill port and the outlet to the pump suction line shall be equipped with a strainer. The reservoir shall be equipped with an easily removed plate to allow for cleaning of the interior. The reservoir shall be vented and done in a manner to preclude rain water or dust from entering through the vent. The pump suction line shall be equipped with a manual valve to stop fluid flow when replacing hydraulic pumps or lines. The hydraulic fluid reservoir shall be common to all circuits of the hydraulic subsystem. The reservoir shall be located so that a positive head shall be provided to all pumps during any operation. Suction lines to hydraulic pumps shall be designed to respond to all fluid flow demands within the capacity of the pumps while maintaining a positive head at the pump intake ports. The capacity of the hydraulic fluid reservoir shall be sufficient to contain all fluid necessary for hydraulic subsystem operation. The hydraulic reservoir shall be of such configuration and located so that any normal operational attitudes of the transporter will not cause the hydraulic head to be lost. The reservoir shall have a tapered bottom to a drain plug for thorough draining. The return line shall end below the minimum operational fluid level to prevent aeration. The reservoir shall be equipped with an easily read and guarded fluid level indicator. The reservoir shall be

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marked with the type hydraulic fluid to be used and any other appropriate warnings. The hydraulic fluid shall be filtered by a return line filter prior to entering the reservoir.

3.7.2 Hydraulic valves. The hydraulic valves shall be grouped in banks to provide easy access for maintenance. The banks shall be installed in weather/debris protection enclosures. All hydraulic valves shall be marked to correlate with the hydraulic schematic. The hydraulic schematic shall be applied to a metal plate and mounted on the valve enclosure(s).

3.7.3 Lift circuits. The hydraulic system shall provide and control power for activation of the deck lifting mechanisms. The circuit(s) shall be capable of flows and pressures required by the operational characteristics of the deck lift mechanisms. The system shall provide for simultaneous and individual raising and lowering of front and rear left mechanisms and shall not require operator proficiency. Check valves shall be mounted at the hydraulic cylinder inlets (pressure port) to preclude an uncontrolled condition in the event of component failure(s). If necessary, restructures shall be installed to control downward descent speed (see 3.9.5).

3.7.4 Hydraulic fluid. The hydraulic fluid shall comply with the requirements of MIL-H-83282.

3.7.5 Hydraulic hoses. Hoses shall be of adequate strength and size to deliver the required hydraulic fluid pressure and volume in accordance with SAE J517. Fittings shall be reusable.

3.7.6 Hydraulic system test fittings. Quick disconnect fittings shall be provided for testing the hydraulic circuits for flow rate and pressure.

3.7.7 Standby operation. A power standby system shall be provided for emergencies to power all circuits in the event of vehicle engine failure. It shall be operable on both internal and external power sources. External power source requirements shall be the same as internal power source capability. External power source receptacle shall be mounted on the side of the vehicle and shall be easily accessible when deck is at any position. The receptacle provided (see 3.5.2) shall serve both functions. Use of internal power source for the standby system shall not require operator assembly or modification of the vehicle and shall be operable when deck is at any position. It shall be possible to perform the following exercise cycle using onboard batteries without rest with the standby system when deck is unloaded without recharge of onboard batteries.

a. Raise and lower front and rear lift mechanisms simultaneously at 2 feet per minute (FPM) throughout entire height range.

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b. Power the deck roll, pitch and side shift functions throughout their entire ranges in less than two minutes each.

c. Provide steering for the vehicle at a reduced rate for 30 minutes.

The standby system shall be electrically activated by a master switch from a switch enclosure mounted on the chassi and shall not be able to be activated when the engine is running. The platform functions will then be controlled by the same electrical switches in the cab used for control of the main hydraulic system.

3.8 Loads.

3.8.1 Full loads. The loader shall be capable of transporting and transferring cargo palletized on military and commercial pallets, ISO containers, tactical shelters, lower deck containers, airdrop platforms and rolling stock, without the use of leveling pads or outriggers.

3.8.1.1 Palletized cargo. Military pallets are the HCU-6/E, HCU-12-E and HCU-10/C per MIL-P-27443. Commercial pallets have standard widths of 60.4, 88 and 96 inches. The 88 inch wide pallets come in lengths of 108 and 125 inches. The 96 inch wide pallets come in lengths of 61.5, 125, 196 and 238.5 inches. The 60.4 inch wide pallet has a length of 125 inches. A full load of palletized cargo shall consist of three HCU-6/E (88 inches x 108 inches) pallets with a total gross weight of 25,000 pounds evenly distributed and have a center of gravity located at 48 inches above the pallet. The three pallets shall be married part of the time and shall be separated the remainder of the time. Individual loaded pallets shall weigh up to 10,000 pounds. The pallets shall be oriented with the 108 inch dimension perpendicular to the longitudinal centerline of the loader.

3.8.1.2 Rolling stock. A full load of rolling stock shall consist of any combination of single axle loads up to 15,000 pounds each and tandem axle loads of up 20,000 pounds for a gross load of 25,000 pounds supported by size 10.00 x 20, 11.00 x 24.5 or 11.00 x 22.5 pneumatic tires within the maximum deck width (108 inches) and a wheelbase of 15 feet.

3.8.1.3 Containerized cargo. Air containers consist of lower deck containers, ISO containers, Container Delivery System (CDS) loads and tactical shelters. The lower deck containers are the LD-1 through LD-11. The ISO containers have an overall height of eight and one-half feet and width of eight feet and are available in lengths of 10 and 20 feet. Specifications for surface intermodal and air/surface intermodal containers are covered by ISO 688 and ISO 8323, respectively. Tactical shelters are of the ISO container configuration and are covered by MIL-STD-907.

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A full load of containerized cargo shall be one fully loaded ISO air/surface container (96 inches wide x 240 inches long x 96 inches high). Gross weight shall be 25,000 pounds.

3.8.1.4 Platform cargo. Military airdrop platform is the Type V platform (per MIL-P-44115). A full platform load shall consist of one 16 foot long, Type V platform loaded to a gross weight of 25,000 pounds and having a center of gravity located at 48 inches above the platform.

3.8.2 Unbalanced loads. The loader shall be capable of transporting and transferring the unbalanced loads listed below with no signs of instability. Raising and lowering speeds of the deck shall not be affected by the unbalanced loads.

3.8.2.1 Partial loads. Any combination of partial loads of cargo specified in 3.8.1, for a gross load of up to 25,000 pounds, shall be positioned anywhere along the deck; e.g., one LD-3 3,500 pound container and one 12 foot Type V 14,000 pound platform.

3.8.2.2 HCU-6/E pallet overloads. Any combination of HCU-6/E pallets loaded up to 15,000 pounds each for a gross load of up to 25,000 pounds shall be positioned anywhere along the deck.

3.8.2.3 Moving loads. Any size rolling stock up to that specified in 3.8.1.2 shall be positioned anywhere along the deck and shall be driven onto, across and then off either end of the deck.

3.9 Deck functions. The deck shall include the following capabilities when loaded as specified (see 3.8). All deck functions shall operate over entire height range when deck is in any pitch, roll or side shift attitude without interference damage or degradation of stability. Transfer of unbalanced, partial and rolling stock loads across the deck shall not result in upward or downward movement of the deck caused by structural or suspension deflection, etc, of more than 3.0 inches.

3.9.1 Powered conveyor. The conveyor system shall be capable of supporting loads (see 3.8). It shall be powered and capable of moving same loads, except CDS loads, at 60 FPM, plus or minus 10 FPM, at governed engine speed in both forward and aft directions when deck is level. Conveyor speeds shall be infinitely variable and controlled from the engine accelerator pedal. System must have the capability of pallet stacking, moving one pallet load at a time when three pallet loads rest on the conveyor. The conveyor shall receive and discharge loads at both ends of the deck. In the event of power failure or malfunctions, the power conveyor system shall allow for one man manual transfer of a single loaded HCU-6/E pallet (10,000 pounds gross weight). Conversion to manual mode shall allow free wheeling of conveyor drive and shall not

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require removal of components. If chain drive is used, adjustment must be provided and, provisions must be provided to maintain tensions on the chains whether loaded or unloaded. Conveyor vertical adjustment to insure adequate friction between cargo loads and powered conveyor shall be provided and shall provide sufficient adjustment to lower conveyor drives below the bottom of the HCU-6/E pallet in the event of conveyor drive damage.

3.9.2 Roll. The deck shall be capable of 4 degrees, plus or minus 0.5 degrees, of roll motion in both the clockwise and counterclockwise directions when measured about the deck longitudinal axis.

3.9.3 Pitch. The deck shall be capable of 6 degrees, plus or minus 0.5 degrees, of pitch motion above and below horizontal.

3.9.4 Side shift. The deck shall have the capability of three inches of lateral motion to each side of the longitudinal centerline.

3.9.5 Elevate. The deck shall have an infinitely variable elevating range, for those loads defined (see 3.8) as measured from the ground to the top of the conveyor, of a minimum height of 39.0 inches for loading C-130 Aircraft to a maximum height of 156 inches for loading DC-8 Aircraft. Under all load conditions, the lift mechanisms shall provide a level raising and lowering speed of 20.0 FPM, plus or minus 5.0 FPM, and provide two level lowering speeds of 20 FPM, plus or minus 5.0 FPM, and 6 FPM, plus or minus 1 FPM. Lift speeds shall be infinitely variable and controlled from the engine accelerator pedal. Under all load conditions, deck levelness shall be controlled by operator manipulation of the deck controls.

3.9.6 Side loading. The loader shall accept side loading of platforms, containers and pallets (see 3.8.1.1, 3.8.1.3, and 3.8.1.4) on the right side of the deck from 463L 10K standard forklifts (MIL-T-52864) or equal standard and adverse terrain forklifts and container handling equipment. The loader shall be capable of accepting ISO containers and shelters (see 3.8.1.3) from top loading container handling equipment; e.g., straddle cranes.

3.9.7 End loading. The loader shall accept front and rear end loading from the A/S32H or equal Trucks, Aircraft Cargo Loading - Unloading. Commercial semitrailers, and rear loading from 463L-10K standard forklifts or equal standard and adverse terrain forklifts of platform containers and pallets (see 3.8.1.1, 3.8.1.3, and 3.8.1.4).

3.9.8 Interface with aircraft. The loader shall be capable of performing all cargo transfer functions at all cargo doors of all

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models of cargo configured C-130, C-141, C-5, C-17, B707 and DC-8 and lower lobe cargo doors of cargo configured B747, B767 and DC10.

3.9.9 Slip rate. The downward slip rate of the elevated cargo deck shall not exceed .250 inch per hour with the engine not running.

3.9.10 Creep rate. The upward creep rate of the deck shall not exceed .250 inch per hour for any elevation, with the engine running and the selector control set for deck lift operation.

3.10 Deck. The loader deck length shall be sufficient to accommodate 22 feet, four inches of cargo described (see 3.8) through the entire elevating range (see 3.9.5). Every portion of the deck located inside the pallet guide rails, except conveyor rollers, shall have the capacity to withstand the imposed forces of rolling stock (see 3.8). The conveyor roller shall be repositioned (e.g., flip-over roller trays) or covered to allow imposed forces of rolling stock and to provide a suitable deck surface for rolling stock. If covers are used, they must be provided with each loader, and provisions for storage shall be provided. Conversion of deck to allow rolling stock loads shall not require more than two manhours. HCU-6/E pallets shall not be used to cover the deck to provide a suitable deck surfaces. When the deck is configured to provide a suitable surface for rolling stock, the deck top surfaces shall be placed in the same horizontal plane, plus or minus 1.0 inch.

3.10.1 Side loading capability. Two tine troughs shall be incorporated into the right side of the deck centered on the deck lateral centerline in order to maximize the available deck space for cargo handling. The troughs shall provide insertion capability below pallets, containers and platforms resting on the conveyor. Tine entry troughs centerlines shall be spaced apart 47.0 inches minimum and 52.0 inches maximum. The minimum opening size shall be 4.0 inches high measured below the A/E29H-1 platform skids, 12.0 inches wide and 78.0 inches long measured to outside edge of deck. There shall be provision for water drainage of tine troughs. The troughs shall be covered by removable decking to preclude personnel injury when not in use. One section of right side guide rail, and entire right side handrail shall be easily removable to allow entry of forklift tines. Removal of section shall expose the two tine troughs.

3.10.2 End loading capability. Two tine entry troughs shall be incorporated into the deck, one on each side of the deck longitudinal centerline, at the rear so as to provide forklift tine insertion capability below cargo pallets, containers and platforms resting on the conveyor. Tine entry trough centerlines shall be spaced apart 47.0 inches minimum and 52.0 maximum. The minimum tine entry opening size shall be 4.0 inches high, 12.0

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inches wide and 78.0 inches long measured to rearmost of rear bumpers. There shall be provision for water drainage. The troughs shall be covered by removable decking to preclude personnel injury when not in use.

3.10.3 Conveyor rollers. At least four rows of rollers shall be located to be compatible with the Type V Airdrop Platform. Rollers shall be so located to provide roller surfaces for the Type V platform roller pads. The Type V platform is described by Army Drawing (FSCM 81337) 11-1-2780. The rollers shall be spaced so as to preclude degradation of loads listed (see 3.8) and shall run full length of cargo deck. Rollers shall be located within three inches of the extreme projection of the loader at both ends. Any remaining structural projection shall be ramped at approximately 45° with no sharp edges.

3.10.3.1 Roller tubes. The roller tubes shall be constructed of aluminum of adequate material and thickness to support transferred loads. The tubes shall be conversion coated per MIL-C-5541, Class 1A. To retain the bearings, the tube ends shall be contributor so as to create the necessary interference when the bearings are pressed in place.

3.10.3.2 Axles. The axles shall be constructed from hexagonal steel rod. The axles shall be cadmium or zinc electroplated in accordance with QQ-P-416, ASTM A123 or ASTM B633, respectively.

3.10.3.3 Bearings. The bearings shall be prelubricated, sealed, hex bored, anti-friction steel ball bearing type, or the equivalent, and adequately rated for the load on the roller. No spacer or retainer is permitted between the roller and the outer race or between the axle and the inner race. The bearings shall be sized and toleranced so as to maintain the necessary interference fit to retain the bearings in place. The bearing housings shall be cadmium or zinc electroplated in accordance with QQ-P-416 or ASTM B633, respectively.

3.10.4 Guide rails. Guide rails shall be steel and shall be configured to provide directional control for 108 inch wide pallets. The guide rails shall run the entire length of the deck. The guide rails shall be positioned 108 inches apart with at least .500 of an inch and no more than .625 of an inch clearance for the pallets. The rail shall be capable of withstanding all forces generated by fully loaded pallets, containers and platforms when the vehicle is operated in any mode required herein. Guide rails shall diverge at each end to aid in entry of cargo onto the deck. Guide rails shall be continuous, except breaks are allowed on the right side for side loading and on the left side for cab relocation during air transport. Vertical edges of rails at breaks shall be tapered to preclude pallet interference.

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3.10.5 Pallet restraints. Pallet restraints shall be incorporated along the guide rails so as to engage a minimum of two notches on each side of HCU-6/E pallets and platforms (see 3.8.1.4) and shall be equally spaced. HCU-6/E pallets shall be either separated or married and they shall be oriented with the 108 inch dimension perpendicular to the longitudinal axis. Allowances shall be provided for the use of pallet couplers (2.0 inches each, 4.0 inches total for a three pallet train). The restraints shall positively lock to eliminate disengagement due to vibration and shall operate by one handed manipulation. The restraints shall be made of stainless steel. The restraints shall be incorporated as part of the guide rail system and be accessible and easily actuated at any deck height position. Restraints shall be capable of withstanding all forces generated by fully loaded pallets and platforms (see 3.8.1) (25,000 pounds) when the vehicle is operated in any mode required herein. It shall not be required to remove or separate pallet restraints from the guide rails for air transport. The restraints shall not at any time or position extend above the top skin of the cargo pallets.

3.10.6 Emergency pallet stops. Two mechanical stop systems (one or two stops per system) shall be provided, one at each end of the deck to stop forward and aft movement of loose pallet/container/platform loads. They shall be located so as to provide a usable deck length of 273.0 inches, plus or minus 1.0 inch. The stops shall be positive, retractable and capable of withstanding repeated impacts from full loads of palletized, containerized and platform cargo (see 3.8.1) moving at a rate of 60.0 FPM without permanent deformation. The stops shall transmit all forces from this impact directly into the frame. Two controls, one front and one rear, shall be provided for individually operating the rear stop system and the front stop system by a one handed, one movement manipulation by an operator standing on the left hand catwalk and on the ground when the deck is five feet above the ground. A single control for simultaneous operation of both stop systems (rear and front) shall not be allowed. Upon contact with pallet/container/platform entering the deck, the pallet stops shall be forced down without damage to the pallet/container/platform and stops and shall automatically return to the stop position when pallet/container/platform is fully positioned on the deck. The stops shall raise and lower using the pallet stop controls with only the operator's assistance and shall positively lock in both the up and down positions. The operator's control shall conform to MIL-STD-1472.

3.10.7 Catwalk. The truck shall include a catwalk along entire length of the deck on the left side of the deck in order to allow adequate access by transportation personnel. The catwalk shall not interfere with the width constraints for air transportability (see 3.1.9). The catwalk shall be capable of supporting a minimum load of 1,500 pounds over the entire length and 300

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pounds per square foot anywhere on the walkway. The catwalk shall be designed to a static safety factor, based on yield strength, of at least two. All walkways and horizontal deck surfaces constituted as walk areas shall be covered with a nonskid coating conforming to MIL-W-5044, Type II, color same as exterior of the vehicle, or shall be constructed of nonskid material. The catwalk shall be no less than 12.0 inches wide and shall be easily removed by two men without the aid of a lifting device for air transport and side loading operations.

3.10.8 Handrails. The handrails shall be provided along the entire length of each side of the deck. The cab may serve as a part of the handrails. The handrails shall be 42.0 inches, plus or minus 3.0 inches, high in accordance with MIL-STD-1472 as measured from the catwalk surface and positioned within 78.0 inches of the longitudinal centerline of the deck. Any access opening shall be protected with a safety chain easily removed when the access opening is to be used. Both handrails shall be easily removed or repositioned by two men without the aid of a lifting device for lower lobe loading, air transport and side loading operation. The handrails shall comply with OSHA standards. Handrails shall be constructed of steel for ease of repair in the field.

3.10.9 Deck access ladder. A self-telescoping, aluminum ladder shall be provided for easy access from the ground to the catwalk (see 3.10.7) at all deck attitudes. The ladder shall comply with OSHA standards.

3.10.10 Bumpers. Bumpers shall be attached to the front, right side and rear of deck to preclude damage during loading/unloading operations to the loader and interface equipment (e.g., docks, other loaders, aircraft).

3.10.11 Tiedown rings. Twelve tiedown rings, six on each side, shall be incorporated as part of the guide rails to provide positive control of pallets, platforms, containers and rolling stock. Six additional low profile or recessed tiedown rings, placed in two rows, shall be provided in the deck floor between the pallet guide rails for positive control of loose parts during air transport. The tiedown rings shall be evenly spaced along the deck length. Each ring and support shall have a capacity of 5,000 pounds minimum in any direction with a static safety factor, based on yield strength, of not less than 1.5.

3.11 Finish and protective coatings.

3.11.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808, except that commercial (see 6.3.2) components may be prepared in accordance with the manufacturer's standard practice, provided it

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is compatible with the exterior finish paint.

3.11.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion or other defect and shall be capable of passing the test described [see 3.11.2.1b.(2)].

3.11.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number of 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be DuPont Imron, or equal. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator griller, bumpers, light trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

NOTE

"Equal" is defined as having performance identical to or superior to DuPont Imron when subjected to the following tests:

a. The following tests shall be in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.

(2) Paragraph 4.3.23 Accelerated weathering.

b. The following tests shall be in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.

(2) Paragraph 4.8.10 Resistance to taping.

(3) Paragraph 4.8.12 Pencil hardness.

3.11.2.2 Desert Sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grills, bumpers, light trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.11.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice.

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3.12 Markings. Unless otherwise specified (see 6.2), marking for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and marking for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.12.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.13 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and be a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.13.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.13.2 Control markings. All controls, valves, gauges, and indicators used in the operation of the vehicle shall be identified by securely attached nameplates installed in clearly visible locations.

3.13.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise, and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.13.4 Lubrication plate. A lubrication plate shall be provided

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directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.13.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instructions for air transport in Types C-130, C-141, C-5 and C-17 Aircraft.

3.14 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Hand tools. The truck shall be designed and constructed in a manner that will allow general maintenance, assembly, and disassembly to be performed to the maximum practical extent with the use of common hand tools. Provide list of special tools and diagnostic equipment (e.g., for microprocessors) necessary to perform field level maintenance in the First Article Test Report.

3.16 Dimensions. The vehicle, in the fully operational condition, shall not exceed 25,000 pounds in curb weight, no more than 27 feet in maximum length and width of the loader will not exceed 150 inches and in air transport configuration overall width will not exceed 111 inches. The height will not exceed 81 inches.

3.17 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in desired level of quality. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

3.17.1 Welders and welding.

3.17.1.1 Welders. All welders employed in the fabrication of

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the vehicle, except for commercial components, shall be certified in accordance with MIL-W-52574. The certification shall be on file at the contractor's facility and available for review by the procuring activity.

3.17.1.2 Welding. Welding procedures shall be in accordance with MIL-W-52574. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.17.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.17.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.17.4 Gear and level assemblies. Gear and level assemblies shall be properly aligned, meshed and operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall not have excessive backlash.

3.17.5 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.18 Operations during aircraft refueling. The loader shall be designed to safely operate concurrently with refueling of aircraft. The loader shall be designed, constructed, and assembled for safe operation in accordance with NFPA 407.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

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4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Transmission oil temperature.

i. Vehicle odometer reading at the start and finish of each test.

j. Engine hourmeter reading at the start and finish of each test.

k. Hydraulic fluid pressure at the pump outlet.

l. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced, and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following

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conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubricant or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

i. Overheating of the engine, transmission, or any other vehicle component.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples shall be identical to the production items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor test sample (see 4.4.2.1).

b. User test sample (see 4.4.2.2).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described (see 4.6.1 through 4.6.9).

4.4.2.2 User test sample. The user test sample shall be subjected to the tests described (see 4.6.1, 4.6.2, 4.6.10) and shall be subject to the tests (see 4.6.1, 4.6.2) prior to the

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test described (see 4.6.10) and shall be shipped to Little Rock AFB until after completion of the first article tests of the contractor test sample (see 4.4.2.1), except the environmental tests (see 4.6.6) and electromagnetic interference (EMI) tests (see 4.6.5.3).

4.4.3 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as part of the test report:

a. All data collected, failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedures, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. List of special tools and diagnostic equipment necessary to perform field level maintenance.

i. Plan for testing production units.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

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4.5.1 Individual testing. Each production vehicle shall be subjected to the tests specified (see 4.6.1, 4.6.2).

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described (see 4.6) with the exception of (see 4.6.3.8, 4.6.5.3, 4.6.5.6, 4.6.5.7, 4.6.6, 4.6.8, 4.6.9 and 4.6.10). Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

4.5.2.1 Rejection and retest. When a vehicle fails to meet the requirements of this specification, vehicles to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. Any painted surface of the exterior of the vehicle may be tested in accordance with paragraph (see 4.8.10) of MIL-C-83286. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Preliminary operation. The fully loaded vehicle shall be driven at various speeds for a minimum distance of 5 miles minimum at an average speed of 10 MPH over improved roads. During the tests, not less than 10 full circle left turns, 10 full circle right turns and 10 sudden stops (12 feet/second^2 minimum) shall be made. The following tests shall be performed on the fully loaded vehicle:

a. Ten cycles of lift, full travel up and down.

b. Ten cycles of pitch, full travel fore and aft.

c. Ten cycles of roll, full travel clockwise and counterclockwise.

d. Run powered conveyor system for 30 minutes.

e. Check engine starting, operation and accessories.

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f. Check all lights (i.e, headlamps, turn signals, work lights, cab lights, etc).

g. Check horn.

h. Check steering with vehicle stationary.

i. Check windshield wipers.

j. Check turn signals.

k. Check all gauges and controls.

l. Check brakes, stationary (i.e, service and parking).

m. Check cab doors and windows.

n. Check climate control system, including defroster.

o. Operate pallet stops/restraints/rollers (powered and nonpowered)/guide rails.

p. Check any miscellaneous equipment.

4.6.3 Mobility tests.

4.6.3.1 Maximum speed and acceleration. The time-speed-distance relationship of the fully loaded vehicle shall be determined from actual acceleration tests from a standing start to maximum speed on a dry level paved surface to demonstrate compliance with [see 3.2.1(a), 3.2.1(b), 3.2.1(h)]. Not less than three runs, each in opposite directions, shall be conducted. Maximum speed and acceleration shall be the average for all runs.

4.6.3.2 Grade operation. The fully loaded vehicle shall be driven up and down a 3.0 percent, dry, paved grade to demonstrate compliance with [see 3.2.1(f)]. The maximum speed that can be maintained while climbing the grade shall be measured and recorded in the test report. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.6.3.3 Side slope operation. The fully loaded vehicle shall be driven in both directions on a side slope to demonstrate compliance with [see 3.2.1(d)]. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the slope.

4.6.3.4 Aircraft ramp test. The unloaded vehicle, when con-

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figured for air transport, shall be driven forward from a hard flat horizontal surface and ascend a ramp with a length of 20 feet and an angle of 15.0 degrees minimum to demonstrate compliance with [see 3.2.1(e)]. At the end of the ramp, there shall be door opening simulating a C-130 Aircraft door to demonstrate compliance with (see 3.1.9) C-130 door clearance. It shall then proceed across the ramp crest onto an upper connecting horizontal surface until all tires are resting on the upper surface and then driven backward down the ramp and onto the lower horizontal surface. The tests shall be repeated in the opposite direction. Compliance to all other aircraft ramp angles shall be demonstrated graphically.

4.6.3.5 Suspension articulation. The vehicle shall be tested to demonstrate compliance with (see 3.2.2). The sides of the blocks may be sloped; however, the block length shall be such that when one tire of a bogie is at the top of the block, the other tire on that side is on the ground.

4.6.3.6 Wall to wall turning diameter measurement. The vehicle wall to wall turning diameter for turns in each direction shall be measured in accordance with SAE J695 to demonstrate compliance with [see 3.2.1(K)]

4.6.3.7 Brake tests.

4.6.3.7.1 Service brake tests. The vehicle shall be tested in accordance with SAE J786a to demonstrate compliance with [see 3.2.1(i)].

4.6.3.7.2 Parking brake tests. The vehicle shall be tested in accordance with SAE J360 to demonstrate compliance with [see 3.2.1(j)].

4.6.3.8 Belgian block test. The fully loaded truck shall be driven 25 miles minimum over Belgian Block at an average speed of not less than 10 MPH to demonstrate compliance with [see 3.2.1(g)]. At least 1/4 miles of continuous Belgian Block road is required per pass. Repeat test when truck is unloaded. Cobblestone road may be used as a substitute for Belgian Block road.

4.6.3.9 Racking test. The loader shall be driven over a course consisting of 3 1/2 inch minimum high obstacles, spaced to provide a racking of the chassis to show compliance with (see 3.2.2). The obstacles cross section shall be a trapezoid. The entrance and exit angles of the obstacles shall be 30 degrees, plus or minus 10 degrees. The distance across the flat of the top surface shall be 12 inches, plus or minus two inches. It is not the object of this test to obtain resonance. The obstacles shall be so staggered that two wheels of same axle will not travel over an obstacle at the same time and shall be so spaced (not more

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than 7.0 feet) that two wheels of the same side will not travel over an obstacle at the same time. The speed shall be 4.0 MPH minimum. Each wheel of the loader shall pass over not less than 2,400 obstacles. If space requires several passes of the course to obtain the required number of obstacles, then the minimum number of obstacles per wheel per pass shall be 24.

4.6.3.10 Dock bump test. The fully loaded truck shall be driven at 2.5 MPH into a permanent loading dock 10 times so as to impact the deck bumpers when deck is raised to six feet above the ground to show compliance with [see 3.1.2(1)]. Repeat test when deck is unloaded.

4.6.3.11 Vehicle cone index. The vehicle shall be tested to demonstrate compliance with [see 3.2.1(o)]. During 10 separate passes in different tracks, the vehicle shall traverse terrain consisting of unflooded fine grained dry to moist soil having a cone index rating in the critical layer of no greater than 90. For test purposes, it is allowed for the actual soil sample Vehicle Code Index (VCI) to be as high as 120. Each pass shall be at least 60 feet in length. The truck shall come to a complete stop approximately midway across during each pass.

4.6.4 Capacity testing.

4.6.4.1 Pallet transfer test. The truck shall be moved to a dock and shall be used to transfer a full load of pallets (see 3.8.1.1), containers (see 3.8.1.3) and platforms (see 3.8.1.4) across the entire length of the deck to show compliance with see (3.9, 3.10.4, 3.10.5, and 3.10.6). At least three runs of conveyor movement with each load shall be performed at maximum speed.

4.6.4.2 Deck lift speed test. Deck shall be raised and lowered to show conformance to 3.9.5. At least three runs at maximum speed shall be performed.

4.6.4.3 Emergency pallet stop impact test. The emergency pallet stops shall be impacted by full loads of pallets, containers and platforms (see 3.8.1) to show conformance (see 3.10.6). At least ten runs of impact with each load at 60 FPM minimum shall be performed.

4.6.4.4 Unbalanced load. A 15,000 pound minimum load (one HCU-6/E pallet overloaded with 15,000 pounds) shall be positioned at the front of the loader deck (see 3.8.2). The loaded deck shall be raised from the lowest to the maximum height and then lowered to the minimum height 25 times. The load shall be moved to the rear end and the test repeated. Check for levelness of deck (see 3.9.5) during lift cycles.

4.6.4.5 Rolling stock test. A test vehicle with 25,000 pounds

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minimum gross weight, two axle weights of 10,000 pounds minimum and 15,000 pounds minimum, shall be rolled or driven across the entire length of the loader deck ten times to show conformance with (see 3.8.1.2). Tire size shall be either 10.00 x 20, 11.00 x 24.5 or 11.00 x 22.5 pneumatic tires.

4.6.5 Design verification.

4.6.5.1 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and the fully loaded conditions. Shall be in compliance with (see 3.1.5). The fully loaded vehicle shall include ballast to simulate a 250 pound operator in each seating position.

4.6.5.2 Dimensional verification. The dimensions specified herein (see 3.1.6 and 3.16) shall be measured with the vehicle standing on a flat level surface. Overall height and vehicle lean shall be measured in both the empty and the fully loaded configurations, as defined (see 4.6.5.1).

4.6.5.3 Electromagnetic interference measurement. The vehicle and all electrical components be tested in accordance with MIL-STD-462 to demonstrate compliance with (see 3.1.7).

4.6.5.4 Engine speed. The engine governed speed shall be checked to show conformance with (see 3.3.1.1). The engine manufacturer's commercial literature shall be provided showing their recommended governed speed.

4.6.5.5 Sound level. The truck shall be tested in accordance with SAE J336a, J366b and J1077 to demonstrate compliance with (see 3.1.8).

4.6.5.6 Structural analysis. Stress analysis for worst case conditions (including shear and moment diagrams and stress and design safety factor calculations) shall be furnished to show conformance with (see 3.1.11). A Goodman analysis of all fasteners with maximum installation torque shall indicate infinite life.

4.6.5.7 Strain gauge testing.

4.6.5.7.1 Test course. For each test cycle, the vehicle shall be driven at 4.0 MPH over the racking course (see 4.6.3.9) with a minimum of 24 obstacles per wheel per pass. Only one pass is required.

4.6.5.7.2 Patterning. High stress areas, as shown in (see 4.6.5.6) of the vehicle frame assembly, lifting mechanisms and deck structure shall be coated with brittle lacquer to determine the proper locations and orientation of the strain gauges. The

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loader shall be fully loaded and driven over the test course at a minimum speed of 4.0 MPH.

4.6.5.7.3 Dynamic strain recording. Strain gauges shall be positioned as indicated by the brittle lacquer patterning at high stress areas of the vehicle frame assembly, lifting mechanisms and deck structure. A minimum of 20 strain gauges shall be used. Strains shall be recorded for each of the three cycles of the test course:

a. Fully loaded, one cycle.

b. Forty to sixty percent of full load, one cycle.

c. No load, one cycle.

Strain/occurrence histograms shall be supplied for each gauge for each test cycle. The actual test course shall be recorded on the histogram. A picture of each gauge, indicating its position, shall be provided.

4.6.5.7.4 Static strain recording. Strain gauges shall be positioned as indicated by the brittle lacquer at high stress areas of the vehicle frame assembly, lifting mechanisms and deck structure. A minimum of 20 strain gauges shall be used. Strain shall be recorded for each of the gauges when the deck is then fully loaded and in a static condition.

4.6.5.7.5 Evaluation. The minimum ultimate tensile and yield strengths of the gauged components shall be documented. Any strain recorded indicating static stress greater than one third of the material minimum yield strength or a dynamic stress greater than one half of the material yield strength shall be cause for rejection.

4.6.5.8 Static overload test. Place 37,500 pounds minimum cargo, evenly distributed over three HCU-6/E pallets on the truck deck for one hour when the deck is positioned above the deck stops, but not more than six inches above same. Remove cargo and inspect structures for permanent deformation.

4.6.5.9 Hydraulic line failure test. While fully loaded and the deck raised five feet above the deck stops, the hydraulic system shall be rigged to permit free flow of hydraulic fluid to simulate a hydraulic line failure in the lift circuit. Under this condition, the lowering rate shall be not more than 1/4 inch per hour (see 3.9.9). The simulated failure shall be as close to the lift cylinder as possible.

4.6.5.10 Aircraft compatibility test. It shall be demonstrated by physical tests or graphics that the loader complies with paragraph (see 3.9.8). For physical tests, actual or simulated

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aircraft of all those specified herein shall be tested. If simulation is utilized, the models shall be complete as necessary to confirm interface capability without interference problems. Drawings of all models of each type aircraft shall be included to verify configuration accuracy of simulation models. For graphics, drawings of all models of each type aircraft with loader drawing superimposed shall be furnished showing interface capability.

4.6.5.11 Stability test. The truck shall be tested to show compliance with (see 3.3.4.2). The truck, when fully loaded and the deck elevated to the top position, shall be driven on a level surface at a speed of at least 1.0 MPH in a forward direction, in a full right hand turn, and then stopped suddenly at a deceleration rate of at least 12 feet/second^2 without loss of stability. Measure the lateral and longitudinal movement. Repeat test in a full left hand turn and repeat test in a full right hand turn moving in a reverse direction. Repeat test in a full left hand turn and moving in a reverse direction. Use calculations to show that the addition of a 40 knot wind load would not cause the truck to be unstable.

4.6.5.12 Slip and creep rate test. Vehicle shall be tested for a minimum of two hours for each slip and creep rates to show compliance with (see 3.9.9 and 3.9.10).

4.6.6 Environmental tests.

4.6.6.1 Low temperature storage and operation. The complete fully loaded vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with [3.1.4(a)]. Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph (see 4.6.6.5). Engine starting aids may be used (see 3.3.1.1.2). A fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within 60 minutes and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the minimum required cold storage temperature of [see 3.1.4(a)] in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the minimum required cold operational temperature of 3.1.4(a) in a steady-state condition.

4.6.6.2 High temperature storage and operation. The complete fully loaded vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensors locations shall be approved by the procuring activity.

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Operational checkouts shall be performed in accordance with 4.6.6.5. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the maximum required hot storage temperature of [see 3.1.4(a)] in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the maximum required hot operational temperature of [see 3.1.4(a)] in a steady-state condition.

4.6.6.3 Rain. The complete fully loaded vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810, to demonstrate compliance with [see 3.1.4(c)]. Operational checkouts shall be in accordance with (see 4.6.6.3). The interior of the cab and all compartments shall be examined for moisture; any evidence of leakage shall be cause for rejection.

4.6.6.4 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with (see 3.1.4).

a. Humidity - Method 507.2, Procedure I.

b. Salt Fog - Method 509.2, Procedure I.

c. Sand and Dust - Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1,750 ft/min and high temperature adjustment shall be 125° F. For Procedure II, air velocity shall be 5,700 ft/min; sand concentration of 0.0623 g/ft^3, plus or minus 0.015 $g.ft^3$; 90 minutes per face duration.

4.6.6.5 Operational checkout. All controls, levers, mechanisms, and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation. These actions are to result in the operation of the following systems, but will not be limited to:

a. Engine starting, operation and accessories; accessory powerplants.

b. Lift mechanism, all attitudes.

c. Pumps; auxiliary pumps.

d. Climate control system, including defroster.

e. Cab doors and windows.

f. Lights (i.e, headlamps, turn signals, gauges/controls,

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work lights, cab lights, etc).

g. Brakes, stationary (i.e, service and parking).

h. Windshield wipers.

i. Drivetrain (distance limited by chamber size).

j. Aircraft interface.

k. Steering.

l. Winches.

m. Hydraulic hoses/electrical lines/connectors.

n. Equipment/compartment covers/doors.

o. Pallet stops/restraints/rollers (powered and nonpowered)/guide rails.

p. Handrails/ladders/etc - removal/installation.

q. Miscellaneous equipment.

r. Deck, all attitudes.

4.6.7 Reliability demonstration. Compliance with the quantitive reliability requirement (see 3.1.3.1) shall be demonstrated by the tests in (see 4.6.7.1, 4.6.7.2, and 4.6.10). Scheduled maintenance, maintenance events, or adjustments authorized by this specification or the procuring activity shall not be counted as failures. A maintenance event is defined as any malfunction that entails maintenance or adjustment, but does not prevent, delay, or reduce performance of the vehicle as specified herein. After completing all tests, visually inspect all structural welds. Welds shall be clean and free of paint. The structure includes the deck, lift mechanisms and chassis.

4.6.7.1 Vehicle durability test. The fully loaded truck shall be driven for 50 uninterrupted laps without rest over a hard surface course and shall maintain an average lap time of not more than 10 minutes. A lap shall be not less than 0.5 mile in length. During each lap, one left full circle turn and one right full circle turn at maximum cramping angles, one sudden stop, and one transfer at a working height of at least 96 inches shall be performed. All restraints shall be engaged, the emergency stops shall be activated, and the deck and chassis (when applicable) shall be positioned in the travel mode when the truck is in motion. A full circle turn at maximum cramping angles is when the truck performs a complete 360 degrees circle within 130 percent of its shortest capable clearance circle diameter

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[see 3.2.1(k)]. A sudden stop shall be when the truck is stopped at a deceleration rate of at least 12 feet/second^2. A transfer shall be when the full load is off loaded from the truck and then on loaded back onto the truck across the deck front end. The pallets shall be coupled together as one load. The load shall be transferred by means of the truck powered conveyor. The second half of testing shall be the same as the first half, except the truck shall be driven in the opposite direction, the loaded pallets shall not be coupled together, and the uncoupled pallets shall be manually off loaded/on loaded across the rear end. Refueling shall not be allowed (see 3.3.7.1). The average lap time shall be calculated as total test time divided by the number of laps performed. Start time, stop time, and number of laps shall be recorded. The right and left full circle turning radiuses and the sudden stops performed shall be measured and recorded for every lap. Repeat test 19 times.

4.6.7.2 Deck and suspension operational test. The fully loaded truck shall be subjected to 25 full uninterrupted deck and suspension systems exercise cycles (see 3.2.3) at full speed without rest. Install at least 12 temperature probes throughout the hydraulic system to measure the hydraulic fluid temperature for determining the hot spots of the system. Temperature probes shall be located at, but not limited to, the following locations:

a. Pump(s) outlet supply line.

b. Reservoir inlet(s) return line.

c. Reservoir midpoint (approximate).

The temperature of the fluid shall not exceed the components manufacturer's recommended operating range or 185° F, whichever is lower. The hydraulic tank shall contain the normal operating volume of fluid and no auxiliary cooling shall be allowed during the test. For the first and last cycle, deck roller height from ground level shall be measured and recorded to show conformance to (see 3.9.5). Record the temperatures of all probes at the end of each exercise cycle. Tabulate the temperature readings to determine if the system had stabilized below 185° F. Check for levelness of deck during raising and lowering. Check for permanent deformation caused from binding, twisting, etc. Repeat test 15 times.

4.6.8 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed (see 3.1.3.2.1) shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks

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shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing daily maintenance checks by operator's wearing arctic mittens shall be demonstrated.

4.6.9 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with (see 3.1.9). The analysis shall include the guidons and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Types C-130 and C-141 Aircraft.

4.6.10 User test. The vehicle shall be delivered to Little Rock AFB AR for 60 days of operational use. The user test sample shall not be shipped to the AFB without prior 15 day notification to the Procurement Contracting Officer (PCO). The test period shall begin after the vehicle has been inspected and prepared for use and shall end when 60 days of availability are complete or 75 calendar days have elapsed, whichever occurs first. If the user test sample was not available for 60 test days after 75 calendar days have elapsed, the user test shall be considered a failure. The vehicle shall be available for any eight hour period specified by the Government each day, seven days a week. The period shall be scheduled by the AFB representative and the schedules shall be provided to the contractor at least 12 hours in advance. Any period of more than two hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day of the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by the Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the vehicle. The preliminary technical data shall contain procedures for preparation / tiedown of the truck for air shipment; however, the loader will not be air transported. The contractor will provide any special tools/ equipment required to perform operators maintenance/scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before notification to the AFB representative, and Government personnel shall be allowed to observe their actions. The contractor shall provide all truck consumables in the course of the test except for diesel fuel, water, compressed air and electrical power which will be furnished by the US Air Force. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in or simulate

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normal Air Force usage and will not exceed any operating parameters defined by Section 3 of this specification. Noncompliance with the performance requirements of this specification shall be cause for rejection. The contractor shall maintain a daily log. The log shall be signed by the local AFB representative and the contractor after each day of activity. The daily log shall document, but not be limited to, the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. AFB representative comments.

The contractor shall include the user test results, including daily logs of activities, in the first article test report. All systems and subsystems may be tested at the beginning and end of the test period. The contractor shall be responsible for recording all data required for the first article test report.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES

6.1 Intended use. This truck intended for use at air terminals to transport cargo such as palletized cargo, containers, wheeled loads and platforms from the storage docks to the military and commercial aircraft and vice versa. This truck has such features as end loading and side loading, powered conveyor, deck side shift, lower lobe loading of commercial wide bodied cargo aircraft, loading of military cargo aircraft and drive on/drive off air transportability of C-130, C-141, C-17 and C-5 Aircraft.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.11.2.1).

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c. Rustproofing, if other than the manufacturer's standard commercial practice, is required (see 3.11.3).

d. Marking, paint required (see 3.12).

e. Identification markings required (see 3.12.1).

f. When required, ten HCU-6/E pallets, and one 16 foot Type V platform shall be loaded by the Government to the contractor (see 3.8.1).

6.3 Definitions.

6.3.1 Belgian block. A Belgian Block course is paved with unevenly laid granite block, forming an undulating surface. It duplicates a rough cobblestone road such as is found in many parts of the world.

6.3.2 Commercial. A manufacturer's standard (product), having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.3.3 Fully loaded truck. Except as otherwise specified, a fully loaded truck shall be the first article test sample loaded with 25,000 pounds, plus two percent, of weight evenly distributed over three 463L HCU-6/E pallets with a center of gravity located at 48.0 inches above the pallet top surface.

6.4 Subject term (key word) listing.

Truck, loader
Aircraft cargo
Loading, unloading
MIL-T-38045 A/S32H-5
Side loading
End loading
Palletized cargo
Containers
Powered conveyor
Diesel, engine

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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